Experiment 13: Write a python program to Minimax Algorithm for Gaming

Aim:

To write a Python program to implement the Minimax algorithm for a two-player game.

Theory:

The Minimax algorithm is used in Artificial Intelligence for decision-making in games such as Tic-Tac-Toe. It has two players:

- MAX: Tries to get the highest score.
- MIN: Tries to get the lowest score.

The algorithm explores possible moves and selects the best move assuming both players play optimally.

Algorithm:

1. Start the program.
2. Define the game tree with possible scores.
3. If the current level is MAX, choose the maximum value.
4. If the current level is MIN, choose the minimum value.
5. Repeat until the final values are reached.
6. Display the best value.
7. Stop the program.

Python Program:

```
# Minimax Algorithm for Gaming

def minimax(depth, node, is_max, scores):
    if depth == 2:
        return scores[node]
    if is_max:
        return max(
            minimax(depth + 1, node * 2, False, scores),
            minimax(depth + 1, node * 2 + 1, False, scores)
        )
    else:
        return min(
```

```
        minimax(depth + 1, node * 2, True, scores),
        minimax(depth + 1, node * 2 + 1, True, scores)
    )
scores = [3, 5, 2, 9]
best = minimax(0, 0, True, scores)
print("Best value using Minimax:", best)
```

exp 13 max algorithm.py - C:/Users/yadal/OneDrive/Documents/AI/exp 13 max algorithm.p...

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```
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scores = [3, 5, 2, 9]

best = minimax(0, 0, True, scores)

print("Best value using Minimax:", best)
```

Ln: 24 Col: 0

Output:

```
>>>
===== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 13 max algorithm.py ====
Best value using Minimax: 3
>>>
```

Result:

Thus, the Minimax algorithm was successfully implemented in Python to select the best possible outcome in a two-player game.